

Regional rare-earth element supply and demand balanced with circular economy strategies

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The growing dependence of society on rare-earth elements poses a challenge to achieving a just low-carbon transition globally. While circular economy strategies have gained attention, their specific impacts remain unmeasured. Here we present an integrated model that quantifies how circular economy strategies can reshape global supply chains of the critical rare-earth elements such as neodymium, praseodymium, dysprosium and terbium. The model considers both in-ground and in-use stocks across ten regions from 2021 to 2050. The projections include the full deployment of three widely accepted climate scenarios. We find a considerable mismatch between in-ground stocks, supply and demand at specific region and element levels, with the mismatch for heavy rare-earth elements as a key obstacle for achieving net-zero emissions targets. We suggest that, as in-ground stocks decline among mineral suppliers, the accumulation of in-use stocks in consuming regions can foster a more balanced and less polarized geopolitical landscape for rare-earth elements, and circular economy strategies can lead to an increase of 701 kt secondary supply and a decrease of 2,306 kt demand within the next three decades. Implementing these circular economy strategies will require international cooperation in the governance of rare-earth elements amid sustainable transition.

Climate change is a common challenge for the entire world, which requires efforts from nearly all nations for a just low-carbon transition^{1,2}. However, the supply of rare-earth elements (REEs), especially neodymium (Nd), praseodymium (Pr), dysprosium (Dy) and terbium (Tb), as critical raw materials of clean technologies (for example, electric vehicles (EVs) and wind power), is highly concentrated in a few countries^{3–6}. This has sparked ‘anxiety’ about supply security and trade friction, such as in the United States⁷ and the European Union⁸.

Furthermore, the uneven distribution of REEs is widely concerning as it may intensify international competition and geopolitical risks and further impede global climate goals, as noted by some international organizations^{9–11} and others^{12,13}. To facilitate the transition towards a climate-safe future, it is essential to develop effective strategies for ensuring the stability of global REE supply chains.

Unlike fossil fuels, which are ‘burnt out’ and permanently lost once consumed, REEs¹⁴ and other metals are accumulated as in-use stocks

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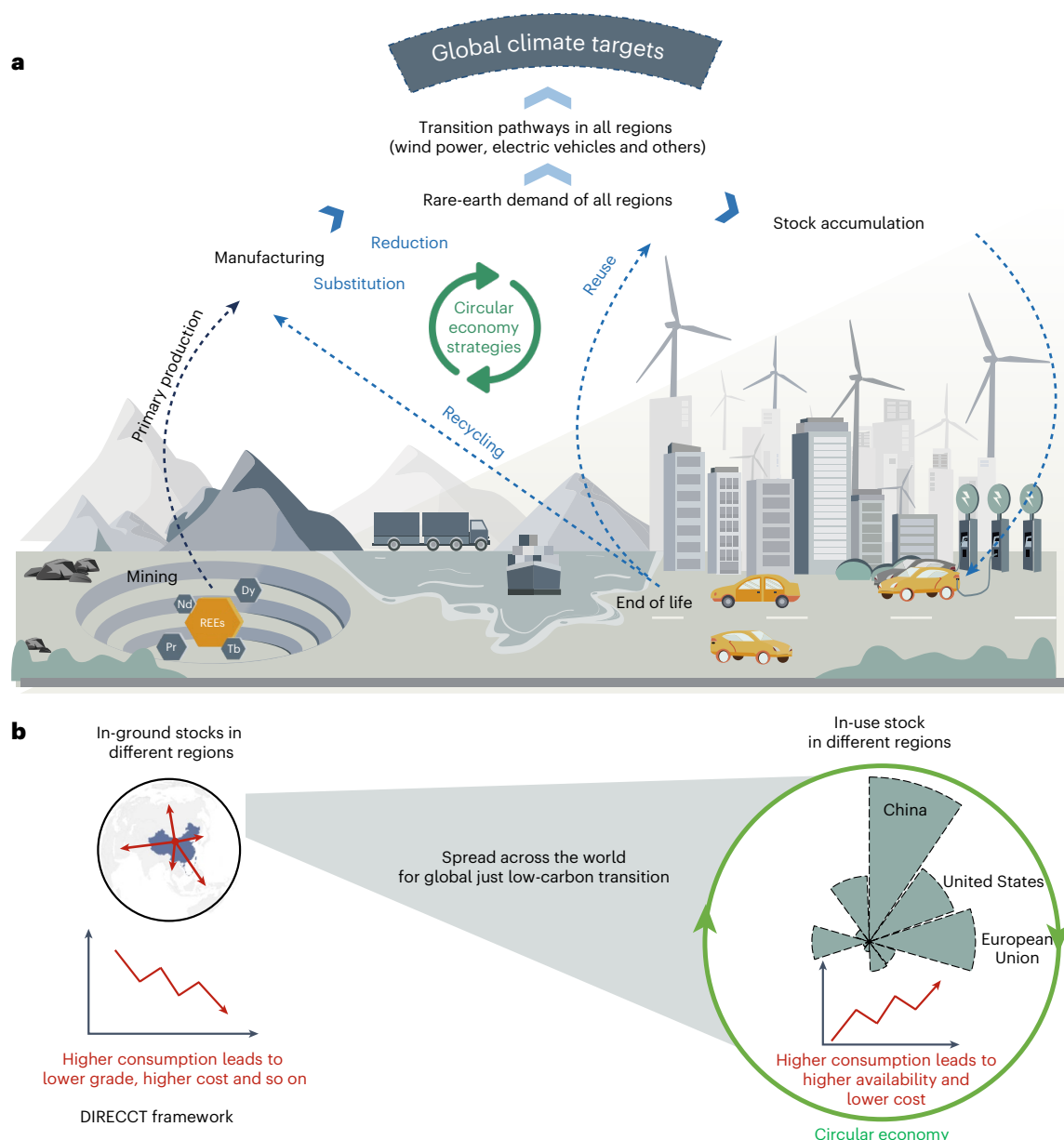


Fig. 1 | Framework of DIRECCT. This framework aims to map the REE flows from in-ground to in-use stocks for global just transition. **a**, The roles of different circular economy strategies, including reuse, reduction, substitution and

recycling, in the shift of in-ground minerals to in-use stocks. **b**, How the transition of REEs can assist the clean and just transition along with its spread from mining sites to final consumers across the world.

that can be ‘recovered’ as alternative supply^{15,16}. Given that massive REEs transfer from in-ground stocks (deposits) to in-use stocks (products), there is growing interest in extracting the in-use stocks as secondary supplies¹⁷. Circular economy (CE) strategies, which have been proposed to reduce supply issues across various critical materials such as cobalt and lithium^{18,19}, can also be applied to the case of REEs^{6,20,21}. However, despite recent attention⁵, the potential of rare-earth (RE) circularity in promoting just low-carbon transition remains uncertain due to limited integrated quantitative analysis.

The present focus from previous literature^{4,22–24} regarding RE geopolitics is limited to the regional holdings of in-ground stocks. By contrast, the corresponding impact of in-use stocks, which refers to REEs in the final products¹⁵, has received scant attention. Importantly, once they are used, traded and diffused globally, REE endowment will be dispersed across the world (Fig. 1b). This can form a new geopolitical chessboard. Most previous studies, from either a single regional

perspective^{5,25,26} or an entire global perspective^{10,18,27}, disregarded the disparities between regions. This may fail to capture the complexity and shifting geopolitical dynamics of the REE market that matter to global just transition.

In this Article, we present a Dynamic Integrated Model of Rare-Earth Circularity and Climate Target (DIRECCT; Fig. 1a) model to explore the linkage among climate targets, energy transition pathways and circular flows of REEs in ten global regions. The DIRECCT model considers three widely adopted climate target scenarios: the stated policies scenario (STEPS), the sustainable development scenario (SDS) and the net-zero emissions by 2050 scenario (NZE). Two types of metal stocks, in Earth’s crust system (in-ground stocks) and in the societal systems (in-use stocks), are captured to quantify the dynamic changes of regional REE distribution (Fig. 1b). In each climate target scenario, the roles of different CE strategies (Fig. 1a) in reshaping global REEs supply for a just and safe low-carbon transition are explored and discussed.

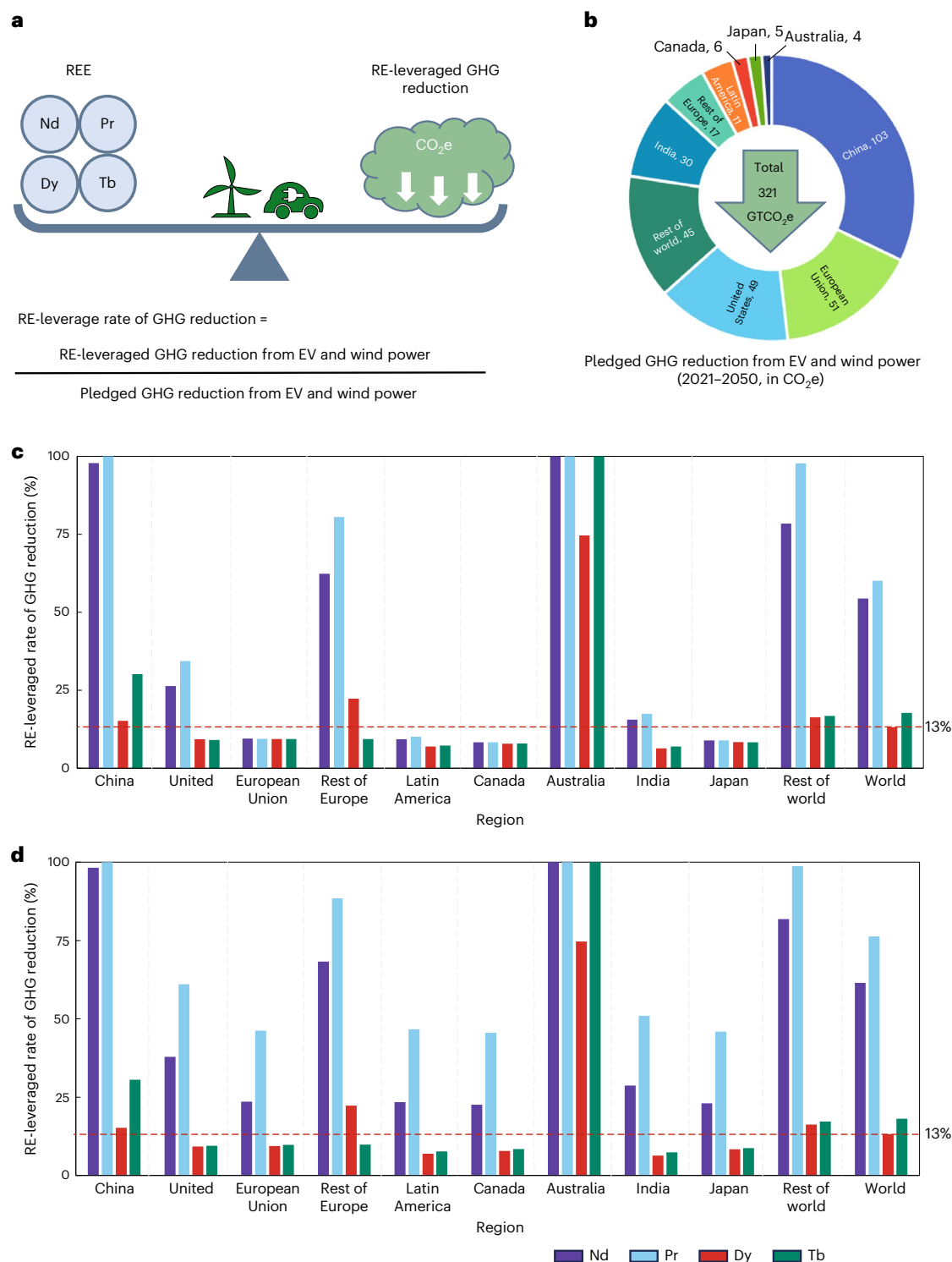


Fig. 2 | RE-leveraged and pledged GHG reduction in the NZE scenario.

a, Sketch diagram of RE-leveraged GHG reduction. **b**, Pledged GHG reduction from EV and wind power from 2021 to 2050 for each studied region (in GtCO₂e).

c,d, RE-leveraged rate of GHG reduction for each studied region under two trade scenarios: counterfactual no-trade scenario (**c**) and free trade with priority in self-supply scenario (**d**).

REEs limiting factor to achieving climate goals

REEs play a crucial role in decarbonizing the economy (Fig. 2a). To reach the NZE target, global greenhouse gas (GHG) reduction from REE-related applications (EVs and wind power) is pledged to be 321 GtCO₂-equivalent (CO₂e) (Fig. 2b). We introduce the indicator ‘RE leverage rate of GHG reduction’ as a proxy to determine the ratio of leveraged GHG reduction supported by available REEs in achieving the pledged climate target

for each region (detailed descriptions in Supplementary Information section 2.4). Considering the uneven RE distribution, we further examine the variations in the RE leverage rate under two trade scenarios (counterfactual no-trade scenario and free trade with priority in self-supply scenario; Supplementary Information section 2.4) in Fig. 2c,d, respectively.

Our results indicate that the ambitious NZE target would be severely hindered by REEs at both global and regional levels if no CE

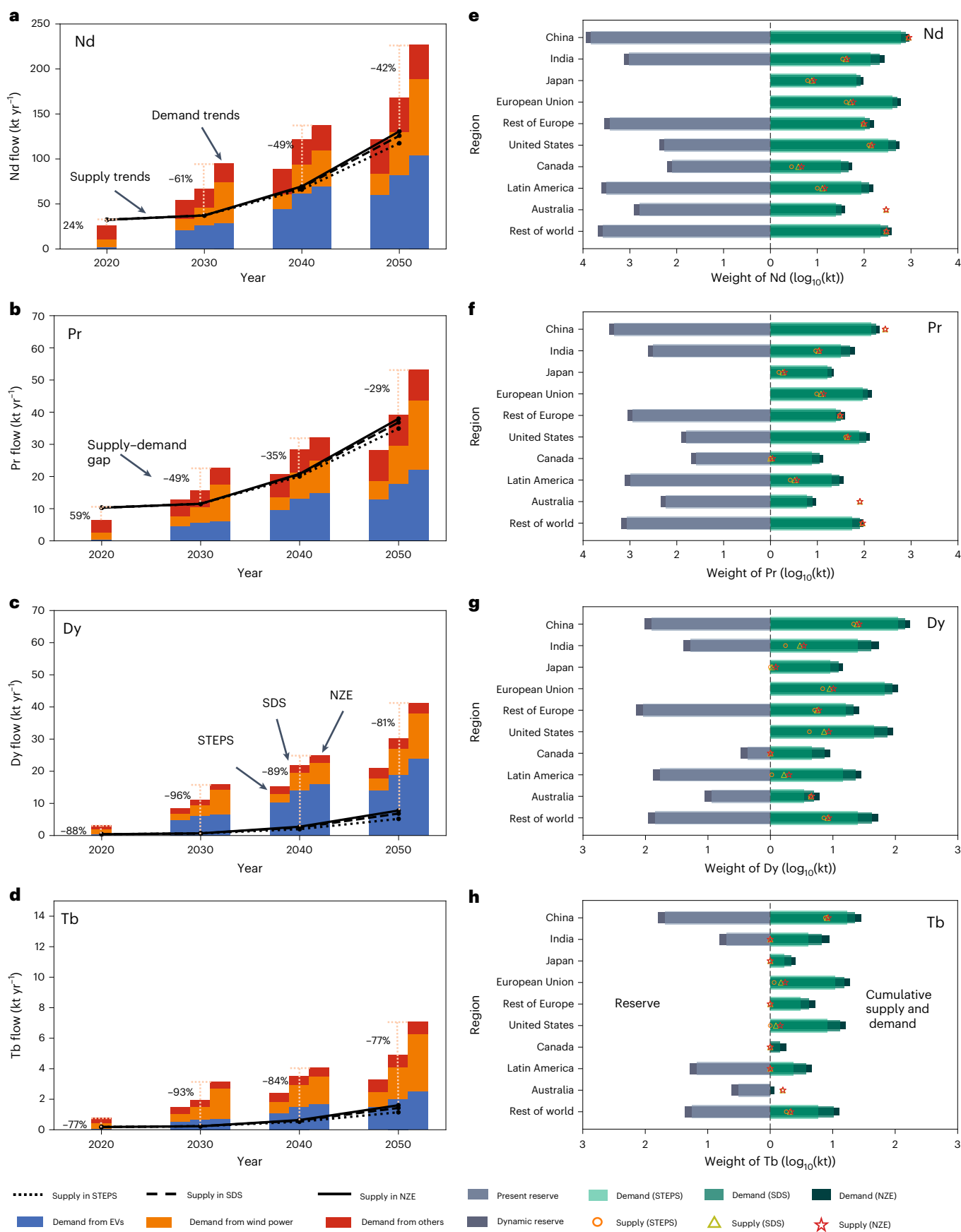


Fig. 3 | The demand, supply and reserve of REEs at global and regional levels in the baseline scenario. a–d, Global (primary plus secondary) supply and demand of Nd (a), Pr (b), Dy (c) and Tb (d) by sector in kt yr⁻¹. **e–h**, Regional reserves, cumulative (primary plus secondary) supply and demand of Nd (e), Pr (f), Dy (g) and Tb (h) between 2021 and 2050, represented as a base-10 logarithm.

strategies were implemented. In the absence of trade, the leverage rates for GHG reduction are as follows: 54% for Pr, 60% for Nd, 13% for Dy and 18% for Tb (Fig. 2c). However, in the 'free trade' scenario, the leverage rates for Pr and Nd can be elevated to 61% and 76%, respectively, while the rates for Dy and Tb remain unchanged (Fig. 2d). Given the attainment of climate target is contingent on all studied REEs, Dy will emerge as the most critical bottleneck raw material. Under the limitation of Dy at the no-trade scenario, China, the European Union and the United States can achieve only 15%, 9% and 10% of their respective pledged GHG reductions. Nevertheless, the global RE-leveraged GHG reduction rate may be limited to 13% (Fig. 2c,d), regardless of RE trade conditions.

The constraints posed by REEs on achieving climate pledges are not equal at the regional level. In the no-trade scenario, the leverage rate of GHG reduction in seven of the ten studied regions—the United States (10%), the European Union (9%), the rest of Europe (9%), Latin America (7%), Canada (8%), India (6%) and Japan (8%)—will lag behind the global average (Fig. 2c). If trade is allowed, the leverage rates of Pr, Nd and Tb for regions such as the European Union, the United States, Latin America, Canada, India and Japan show a notable increase (Fig. 2d). However, the limited availability of Dy continues to undermine these efforts. Thus, there is an urgent coordinated need for a more sufficient and balanced supply of REEs.

Regional shortages of REEs

We compare the regional demand of REEs (details in Methods) with their potential supply, including both primary (from minerals, with projected increase in different regions from literature²⁸) and secondary (from end-of-life (EOL) flows) route, in Fig. 3a–d. The regional comparison between the cumulative demand of future REEs and their present reserves is given in Fig. 3e–h. Notably, the present reserve is adopted here to illustrate the gap between cumulative supply and demand, emphasizing the need for future reserve exploration and development of mining capacities, rather than the exhaustion of minerals.

Our analysis covers the period from 2010 to 2050 (negligible REE consumption of studied applications before 2010). Historically, the corresponding demand for REEs has increased rapidly from 17 kt yr⁻¹ in 2010 to 37 kt yr⁻¹ in 2020 driven by the global wind power and EV expansion. In line with the climate targets, global demand for Nd, Pr, Dy and Tb will increase rapidly from 26 kt yr⁻¹, 7 kt yr⁻¹, 3.2 kt yr⁻¹ and 0.8 kt yr⁻¹ in 2020 to 122–227 kt yr⁻¹, 28–53 kt yr⁻¹, 21–41 kt yr⁻¹ and 3–7 kt yr⁻¹ in 2050, respectively (Fig. 3a–d). However, our results indicate the designed expansion of capacity in light REEs (Pr and Nd) supply will soon become insufficient (Fig. 3a,b). The supplies of heavy REEs (HREEs, Dy and Tb) are much scarcer and have already been deficient since 2020, with a supply deficit ratio of 87% for Dy and 74% for Tb, indicating some unidentified sources at present (Fig. 3c,d). This requires a more radical increase in REE supply, especially to deliver the NZE climate goal, which requires 65% and 21% more REEs than the STEPS and SDS, respectively (Supplementary Table 21).

Such an REE shortage becomes more severe at regional levels (Fig. 3e–h). Among the leading global REE consumers—China, the European Union and the United States—China is projected to be the only nation capable of fulfilling its domestic demand of Pr, Nd and Tb (922 kt of Nd, 214 kt of Pr, 173 kt of Dy and 28 kt of Tb in the NZE) with its present reserve, while the present reserve of Dy is insufficient for all regions. The United States will fall far short of satisfying its domestic demand (558 kt of Nd, 131 kt of Pr, 93 kt of Dy and 16 kt of Tb), even if considering dynamic changes in reserve exploration (Methods). Both the European Union and Japan have almost no domestic reserves, exhibiting profound reliance on the global REE market. Regarding REE supply, China and Australia emerge as the sole sources, with a surplus supply of light REEs (Fig. 3b,d), while the scarcity of HREEs is widespread, with only Australia demonstrating a marginally adequate supply of Tb (Fig. 3h). As a result, the inevitable prospect of international competition for available REEs, particularly HREEs, looms large.

Redistribution of in-use REE stocks

The RE minerals are not equally distributed, with 87% in the top four nations (Fig. 4a). Our projection indicates around 71% of in-use stocks will be accumulated in the top three nations—China (32%), the European Union (21%) and the United States (18%)—by 2050 (Fig. 4b). Regional RE holdings in the form of in-ground and in-use stocks are measured in economic value²⁹ (details in Supplementary Information section 2.5). Notably, the European Union, the United States and Japan are predicted to accumulate US\$169 billion, US\$143 billion and US\$23 billion of RE in-use holdings by 2050, which far exceed their present in-ground holdings (negligible for the European Union and Japan, US\$44 billion for the United States; Fig. 4a,b). This implies that these regions can leverage these to offset their deficiencies of in-ground minerals.

Global present in-ground stocks of REEs are predicted to decrease by 8,070 kt from 2020 to 2050 (Fig. 4c–f). At the same time, there is a notable increase in RE in-use stocks, growing from 219 kt in 2020 to 3,252 kt in 2050. In total, the stocks of studied REEs will decrease from 24,700 kt (US\$4,557 billion) in 2020 to 19,660 kt (US\$3,182 billion) in 2050 due to large REE losses along their life cycle³⁰. This highlights the need for other supply-side strategies in reducing the production loss rates and mining from tailings and slags. Still, the in-use stocks of Nd and Pr will account for 15% and 11% of their total holdings, but the present reserves of Dy and Tb would be exhausted by around 2035–2045 to meet the NZE climate target (Fig. 4e,f). Given that reserve is a dynamic concept, our results regarding reserve exhaustion should be carefully interpreted under the strict condition of no discovery of new RE reserves (Methods). Therefore, the in-use stocks of HREEs are of great significance in international competition since their in-ground stocks are diminishing sharply.

Without CE strategies, most of the in-use stocks will end up in waste streams and cannot be used as a future resource. In this Article, we follow similar concepts from previous studies^{19,31} to develop six different CE scenarios (baseline, recycling, reuse, substitution, reduction and composite with detailed assumptions in Methods and Supplementary Table 15) to explore the impact of CE in different regions in Fig. 5 (regional trends in Supplementary Fig. 27).

We map regional material flows of Nd and interregional dependencies to support the transition under the NZE target in Fig. 5a (cumulative flows during 2021–2050; other REEs in Supplementary Fig. 32), in which the CE baseline and composite scenarios are compared. There are three key major benefits from CE strategies. First, on the demand side, CE can nearly halve the required Nd demand from 3,258 kt to 1,687 kt to achieve the same GHG reduction target. Second, aside from reduction in EOL losses, the circularity of Nd can be largely enhanced with an additional supply of 492 kt of Nd, 1.7 times the 288 kt in the baseline scenario. Third, with the assistance of demand reduction and improvements in circularity, CE can largely reduce the global and regional dependency by around 60% on primary RE minerals (with high geopolitical and environmental risks³²), the amount of which will decrease from 2,970 kt to 1,195 kt during 2021–2050.

The impacts vary among different CE strategies (Fig. 5b). As the preventive measures, the substitution and reduction strategies could reduce global REE primary demand by 1,637 kt and 961 kt, respectively (Fig. 5b). As the EOL measure, recycling can help boost the cumulative secondary supply to 932 kt, higher than that of the reuse strategy (606 kt in total). Compared with the baseline scenario, about 60% of Nd, 59% of Pr, 67% of Dy and 63% of Tb could be exempt from being mined underground by the composite strategy. This can help to conserve the scarce HREEs and ensure that their present reserves can be sufficient to meet their demand before 2050 (Supplementary Tables 19 and 21).

The CE can also benefit a more balanced regional REE supply. As indicated by the Herfindahl–Hirschman Index (HHI; Supplementary Information section 2.6), the concentration of global supply could be reduced by up to 15% for Nd, 14% for Pr, 18% for Dy and 27% for Tb by 2050 (Fig. 5c). Around 30–40% of the demand for REEs in China, the

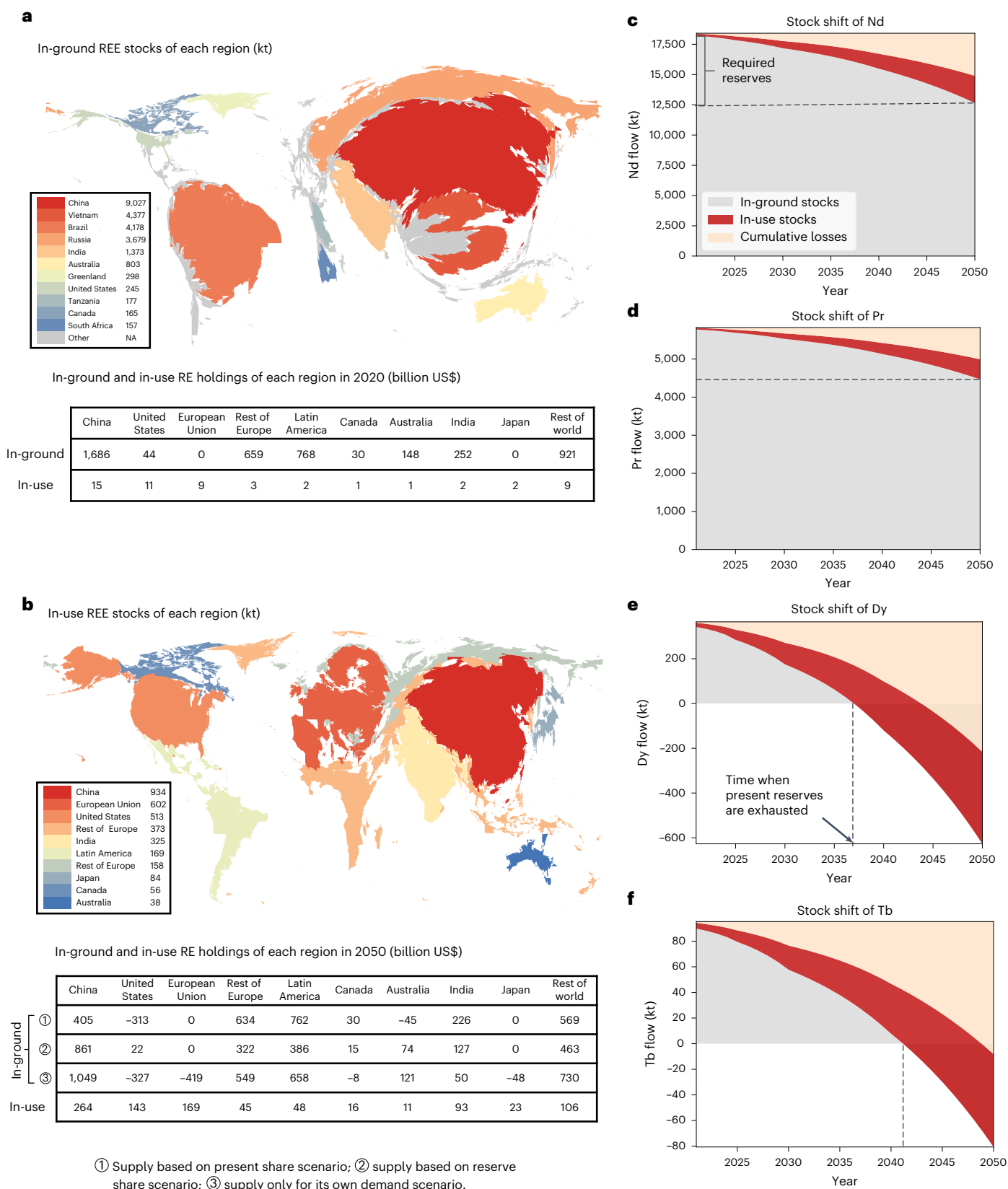


Fig. 4 | Global shift of REEs from in-ground stocks to in-use stocks in the NZE scenario. a, Total in-ground RE stocks and RE holdings of each region in 2020. **b**, Total in-use RE stocks and RE holdings of each region in 2050 (note: rest-of-world countries are coloured on the basis of the average value due to their

loose geographical relationship). **c–f**, REE stocks of Nd (**c**), Pr (**d**), Dy (**e**) and Tb (**f**) shift from in-ground to in-use. The results of STEPS and SDS can be found in Supplementary Information section 3.5.

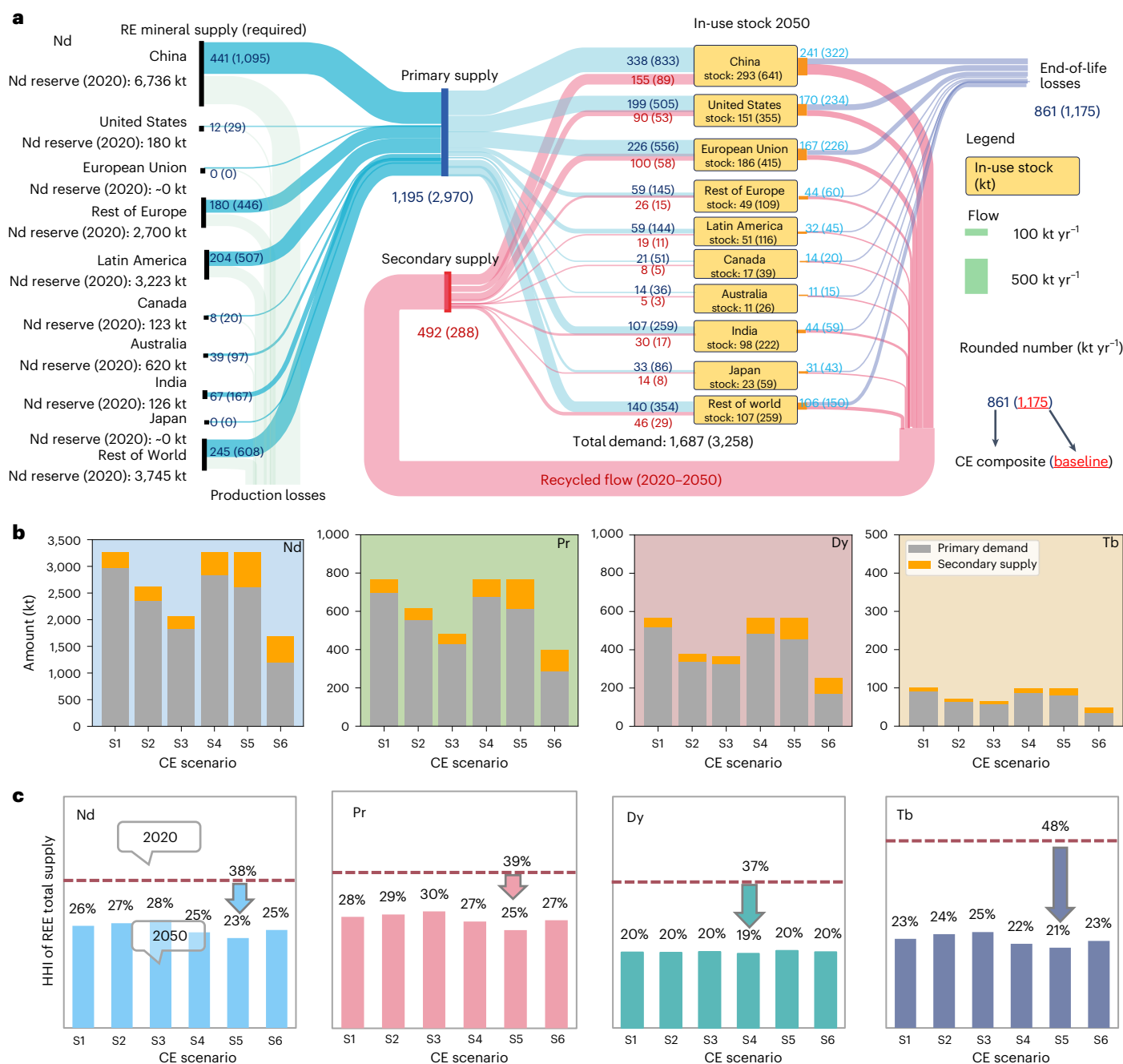


Fig. 5 | Material flows of the REEs and regional interdependence in the NZE scenario under different CE scenarios. a, Sankey diagram of material flows of Nd through the world between 2021 and 2050 in the CE composite and NZE baseline scenarios (for other REEs, see Supplementary Fig. 32). **b**, Global cumulative primary demand and secondary supply of the REEs between 2021 and

2050 (in kt). **c**, Changes in regional concentration of REE supplies, indicated by the HHI of Nd, Pr, Dy and Tb in 2020 and 2050. S1, baseline scenario; S2, reduction scenario; S3, substitution scenario; S4, reuse scenario; S5, recycling scenario; S6, composite scenario.

United States and the European Union could be met through reuse or recycling strategies by 2050 (Supplementary Fig. 27). This is particularly meaningful for the United States and the European Union, which rely highly on foreign REE sources. However, due to the long lifespan of EVs and wind turbines (Supplementary Fig. 27), it is necessary to accelerate reduction and substitution strategies to cope with RE supply risks in the short to medium term.

Under the influence of this single CE strategy, a large amount of foreign RE supply will still be required by the European Union (18–41 kt), the United States (9–30 kt), India (15–33 kt) and Latin America (7–15 kt) to achieve their NZE targets in 2050 (Supplementary Fig. 27). Therefore,

the joint implementation of all CE strategies is a must, which can almost enable China, the European Union and the rest of Europe to eventually achieve a ‘closed-loop’ system (in which the circular flows from EOL REE products will exceed their final demand) of REEs by 2050 in the STEPS and SDS (Supplementary Figs. 25 and 26).

Implementation of CE strategies

The REEs play an important role in fulfilling the global common objective of climate change mitigation. Our investigation reconfirms that the availability of REEs will become an important constraint of global low-carbon transitions, especially the HREEs (Dy and Tb), which could

limit the global GHG reduction to only 13% of its target (Fig. 2). Therefore, the development of HREE-free technologies is a critical and urgent necessity for related industries such as EVs and wind power.

There is growing attention from international bodies to strengthening the governance of critical minerals for global just transition^{33–35}. This is particularly relevant for REEs since their in-ground stocks are considered to be accompanied by high geopolitical, economic, environmental and social risks^{36–38}. Our framework (Fig. 1) can fill a gap in modelling the potentials of CE in reducing the dependence of RE mineral extraction. We show that ongoing consumption of REEs can substantially reallocate REE stocks from their origins to regions with ambitious climate goals (for example, China, the European Union and the United States together accumulate 71% of the world's in-use stocks by 2050; Fig. 4b). Through CE strategies, the newly formed REE stocks can change the geopolitical landscape behind REE supply chain towards a more balanced and less polarized one.

Compared with in-ground stocks, the supply from in-use stocks has many exceptional attributes for a just transition. For example, high demand (price) can incentivize RE mining in more environmentally and socially sensitive areas³³, accompanied by artisanal activities³⁹. Despite demand reduction (halved by CE), the growing supply from in-use stock holders with a more balanced structure (HHI decreased by 14–27%) can help to stabilize REE prices. Meanwhile, secondary production in general has a much smaller environmental footprint than primary supply⁴⁰. Notably, the quality of RE minerals is inevitably decreasing with the depletion of high-grade ore deposits³¹, which further highlights the mentioned advantages of in-use stocks.

However, without CE, the in-use stocks of REEs could not be effectively mobilized. CE is already recognized as the ‘third pillar’ of deep decarbonization^{41,42}. Our results show that CE can not only bring about more complete use of EOL REEs, but also reduce the demand for REEs through both supply-side and demand-side strategies (Fig. 5). For example, CE can substantially reduce primary mining by 60%, which can prevent the ‘REE balance problem’ by reducing the surplus and corresponding cost of other REEs such as cerium and lanthanum^{10,43}. Moreover, CE can help some major regions such as the United States, the European Union, India and Japan reduce their dependency on foreign RE supplies by up to 100%, 93%, 91% and 88% by 2050, respectively (Supplementary Fig. 27)⁴⁴. Thus, international cooperation on CE, such as the European Union's initiative of the Global Alliance on Circular Economy and Resource Efficiency^{45,46}, is welcomed.

The benefits of CE require mobilization of technologies, policies and infrastructures towards full use of secondary REE sources. There is currently a ‘window of opportunity’ when demands and potential tensions are still relatively lower than the anticipated challenges in upcoming decades. In particular, the CE strategies for REEs are associated with some uncertainties^{47,48}. Moreover, the market of RE-containing discards will become more challenging because REEs are highly dispersed and mixed among products^{43,49}. Thus, cooperation on establishing standards between upstream and downstream industries is needed. Notably, the industrialization of CE strategies is also driven by market forces; within this context, enterprises may not attach enough importance to CE in the current window of opportunity. Therefore, it is necessary to closely monitor the development of CE strategies and introduce government regulations and economic incentives to encourage technological innovation and business investment⁵⁰.

Among the limitations in our study (see ‘Uncertainty and sensitivity analysis’ in Methods), the analysis of supply shortage and reserve exhaustion should be carefully interpreted. Indeed, the amount of reserves can increase due to factors such as explorations, technologies and economic developments⁵⁰, such as the new RE deposit in Turkey⁹ or unidentified REE reserves in Myanmar. To fill the supply gap, the importance of REE exploration is still of utmost importance. Nevertheless, the historical growth speed of RE reserves seems not fast, up from 110 million to 120 million tons in the past decade (Methods). Meanwhile,

it usually takes one to two decades from the initial exploration to the operational mining for those newly discovered deposits. In this regard, the time from formation to the recovery of in-use stocks is generally less than 20 years, which highlights benefits in investing CE strategies. Still, a parallel consideration of the exploration of new REE deposits as well as the implementation of CE strategies is recommended.

Online content

Any methods, additional references, Nature Portfolio reporting summaries, source data, extended data, supplementary information, acknowledgements, peer review information; details of author contributions and competing interests; and statements of data and code availability are available at <https://doi.org/10.1038/s41561-023-01350-9>.

References

- Robinson, M. & Shine, T. Achieving a climate justice pathway to 1.5°C. *Nat. Clim. Change* **8**, 564–569 (2018).
- Carley, S. & Konisky, D. M. The justice and equity implications of the clean energy transition. *Nat. Energy* **5**, 569–577 (2020).
- Sovacool, B. K. et al. Sustainable minerals and metals for a low-carbon future. *Science* **367**, 30–33 (2020).
- Li, J. S. et al. Critical rare-earth elements mismatch global wind-power ambitions. *One Earth* **3**, 116–125 (2020).
- Fishman, T. & Graedel, T. E. Impact of the establishment of US offshore wind power on neodymium flows. *Nat. Sustain.* **2**, 332–338 (2019).
- Binnemans, K., McGuinness, P. & Jones, P. T. Rare-earth recycling needs market intervention. *Nat. Rev. Mater.* **6**, 459–461 (2021).
- Building Resilient Supply Chains, Revitalizing American Manufacturing, and Fostering Broad-based Growth* (The White House, 2021); <https://www.whitehouse.gov/wp-content/uploads/2021/06/100-day-supply-chain-review-report.pdf>
- Critical Raw Materials Resilience: Charting a Path Towards Greater Security and Sustainability* (European Commission, 2020); <https://eur-lex.europa.eu/legal-content/EN/TXT/PDF/?uri=CELEX:52020DC0474&from=EN>
- The Role of Critical Minerals in Clean Energy Transitions* (IEA, 2021); <https://www.iea.org/reports/the-role-of-critical-minerals-in-clean-energy-transitions>
- Gielen, D. & Lyons, M. *Critical Materials for the Energy Transition: Rare Earth Elements* (IRENA, 2022); <https://www.irena.org/Technical-Papers/Rare-Earth-Elements>
- Hund, K., Porta, D. L., Fabregas, T. P., Laing, T. & Drexhage, J. *Minerals for Climate Action: The Mineral Intensity of the Clean Energy Transition* (World Bank, 2020).
- Gulley, A. L., Nassar, N. T. & Xun, S. China, the United States, and competition for resources that enable emerging technologies. *Proc. Natl Acad. Sci. USA* **115**, 4111–4115 (2018).
- Akcil, A., Sun, Z. & Panda, S. COVID-19 disruptions to tech-metals supply are a wake-up call. *Nature* **587**, 365–367 (2020).
- Du, X. & Graedel, T. E. Global in-use stocks of the rare earth elements: a first estimate. *Environ. Sci. Technol.* **45**, 4096–4101 (2011).
- Gerst, M. D. & Graedel, T. E. In-use stocks of metals: status and implications. *Environ. Sci. Technol.* **42**, 7038–7045 (2008).
- Chen, W. Q. & Graedel, T. E. In-use product stocks link manufactured capital to natural capital. *Proc. Natl Acad. Sci. USA* **112**, 6265–6270 (2015).
- UNECE. *Guidance for the Application of the United Nations Framework Classification for Mineral and Anthropogenic Resources in Europe* (UNECE, 2022); <https://unece.org/circular-economy/press/united-nations-framework-classification-resources-guidance-europe-gains>
- Heidrich, O. et al. LAYERS: a decision-support tool to illustrate and assess the supply and value chain for the energy transition. *Sustainability* **14**, 7120 (2022).

19. Baars, J., Domenech, T., Bleischwitz, R., Melin, H. E. & Heidrich, O. Circular economy strategies for electric vehicle batteries reduce reliance on raw materials. *Nat. Sustain.* **4**, 71–79 (2021).
20. Schulze, R. & Buchert, M. Estimates of global REE recycling potentials from NdFeB magnet material. *Resour. Conserv. Recycl.* **113**, 12–27 (2016).
21. Geng, Y., Sarkis, J. & Bleischwitz, R. How to build a circular economy for rare-earth elements. *Nature* **619**, 248–251 (2023).
22. Hoenderdaal, S., Espinoza, L. T., Marscheider-Weidemann, F. & Graus, W. Can a dysprosium shortage threaten green energy technologies? *Energy* **49**, 344–355 (2013).
23. Zhou, B., Li, Z. & Chen, C. Global potential of rare earth resources and rare earth demand from clean technologies. *Minerals* **7**, 203 (2017).
24. Grandell, L. et al. Role of critical metals in the future markets of clean energy technologies. *Renew. Energy* **95**, 53–62 (2016).
25. Yao, T., Geng, Y., Sarkis, J., Xiao, S. & Gao, Z. Dynamic neodymium stocks and flows analysis in China. *Resour. Conserv. Recycl.* **174**, 105752 (2021).
26. Imholte, D. D. et al. An assessment of US rare earth availability for supporting US wind energy growth targets. *Energy Policy* **113**, 294–305 (2018).
27. Deetman, S., Pauliuk, S., van Vuuren, D. P., van der Voet, E. & Tukker, A. Scenarios for demand growth of metals in electricity generation technologies, cars, and electronic appliances. *Environ. Sci. Technol.* **52**, 4950–4959 (2018).
28. *Rare Earths Outlook to 2030* 20th edn (Roskill Information Services, 2021).
29. Zeng, X. L., Ali, S. H., Tian, J. P. & Li, J. H. Mapping anthropogenic mineral generation in China and its implications for a circular economy. *Nat. Commun.* **11**, 1544 (2020).
30. Charpentier Poncelet, A. et al. Losses and lifetimes of metals in the economy. *Nat. Sustain.* **5**, 717–726 (2022).
31. Zeng, A. et al. Battery technology and recycling alone will not save the electric mobility transition from future cobalt shortages. *Nat. Commun.* **13**, 1341 (2022).
32. Law, Yao-Hua Politics could upend global trade in rare earth elements. *Science* **364**, 114–115 (2019).
33. Narula, K. et al. *Ensuring Sustainable Supply of Critical Minerals for a Clean, Just and Inclusive Energy Transition* (CSEP, 2023); https://t20ind.org/wp-content/uploads/2023/05/T20_PolicyBrief_TF4_CriticalMinerals.pdf
34. Sturman, K., Loginova, J., Worden, S., Matanzima, J. & Arratia-Solar, A. *Mission Critical: Strengthening Governance of Mineral Value Chains for the Energy Transition* (EITI, 2022); <https://eiti.org/sites/default/files/2022-10/EITI%20Mission%20Critical%20Report%202022.pdf>
35. *Aligning Critical Raw Materials Development with Sustainable Development* (United Nations, 2023); https://sdgs.un.org/sites/default/files/2023-06/2023%20Policy%20Brief%20Aligning%20CRMs-062823_0.pdf
36. Nassar, N. T., Du, X. & Graedel, T. E. Criticality of the rare earth elements. *J. Ind. Ecol.* **19**, 1044–1054 (2015).
37. Golroudbary, S. R., Makarava, I., Kraslawski, A. & Repo, E. Global environmental cost of using rare earth elements in green energy technologies. *Sci. Total Environ.* **832**, 155022 (2022).
38. Ali, S. H. Social and environmental impact of the rare earth industries. *Resources* **3**, 123–134 (2014).
39. Packey, D. J. & Kingsnorth, D. The impact of unregulated ionic clay rare earth mining in China. *Resour. Policy* **48**, 112–116 (2016).
40. Chowdhury, N. A. et al. Sustainable recycling of rare-earth elements from NdFeB magnet swarf: techno-economic and environmental perspectives. *ACS Sustain. Chem. Eng.* **9**, 15915–15924 (2021).
41. Pauliuk, S. et al. Global scenarios of resource and emission savings from material efficiency in residential buildings and cars. *Nat. Commun.* **12**, 5097 (2021).
42. Zhong, X. et al. Global greenhouse gas emissions from residential and commercial building materials and mitigation strategies to 2060. *Nat. Commun.* **12**, 6126 (2021).
43. Fujita, Y., McCall, S. K. & Ginosar, D. Recycling rare earths: perspectives and recent advances. *MRS Bull.* **47**, 283–288 (2022).
44. Nassar, N. T. et al. Evaluating the mineral commodity supply risk of the US manufacturing sector. *Sci. Adv.* **6**, eaay8647 (2020).
45. *Concept Note: Global Alliance on Circular Economy and Resource Efficiency (GACERE): Towards Just Transitions* (UNEP, 2021); https://wedocs.unep.org/bitstream/handle/20.500.11822/40298/GACERE_ConceptNote.pdf?sequence=1&isAllowed=y
46. Wang, P., Wang, H. M., Chen, W. Q. & Pauliuk, S. Carbon neutrality needs a circular metal–energy nexus. *Fundam. Res.* **2**, 392–395 (2022).
47. Elshkaki, A. & Graedel, T. E. Dynamic analysis of the global metals flows and stocks in electricity generation technologies. *J. Clean. Prod.* **59**, 260–273 (2013).
48. Pathak, A. K. et al. Cerium: an unlikely replacement of dysprosium in high performance Nd-Fe-B permanent magnets. *Adv. Mater.* **27**, 2663–2667 (2015).
49. Babbitt, C. W., Althaf, S., Rios, F. C., Bilec, M. M. & Graedel, T. E. The role of design in circular economy solutions for critical materials. *One Earth* **4**, 353–362 (2021).
50. Rauch, J. N. Global mapping of Al, Cu, Fe, and Zn in-use stocks and in-ground resources. *Proc. Natl Acad. Sci. USA* **106**, 18920–18925 (2009).

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Methods

This section provides an overview of our DIRECCT method, and more details can be found in Supplementary Information sections 1 and 2.

The model framework

DIRECCT is a systemic model that combines a stock-driven dynamic material flow analysis (MFA)⁵¹ and a scenario analysis (approach (Supplementary Fig. 1)). It is designed to quantify the future material flows and spatial distribution dynamics of four REEs (Nd, Pr, Dy and Tb) by end use and by region and to evaluate the role that CE strategies play in mitigating REE supply issues. The end uses of REEs can be classified into emerging sectors that deliver climate goals and low-carbon futures, including EV and wind power, and other end uses, for example, robots, elevators and air conditioners. We split the world into ten regions—China, the United States, the European Union, the rest of Europe, Latin America, Canada, Australia, India, Japan and the rest of world—which are indexed by i ($i = 1, \dots, 10$). The results shown in our study are in the period of 2020–2050, in which the time is indexed by t .

Dynamic stock-driven MFA model

A dynamic stock-driven MFA model is used to quantify the annual flows of REE-containing end uses (EVs in units or wind power capacity in GW), which is shown as follows:

$$\text{inflow}_{i,t} = \text{stock}_{i,t} - \text{stock}_{i,t-1} + \text{outflow}_{i,t} \quad (1)$$

$$\text{outflow}_{i,t} = \sum_{m < t} \text{inflow}_{i,m} \times L(t - m) \quad (2)$$

where $\text{stock}_{i,t}$ is the stocks of EVs (Supplementary Fig. 4) or wind power capacity (Supplementary Fig. 5), $\text{inflow}_{i,t}$ is the annual demand for EVs or newly installed wind power capacity, $\text{outflow}_{i,t}$ is the amount of retired EVs or decommissioning wind power capacity, L is the lifetime probability distribution function of EVs or wind turbines, and $L(t - m)$ is the probability of the EVs or wind turbines entering the market in time m and retiring in time t , assuming that the lifespan of EVs follows a Weibull probability distribution with the most likely lifetime of 15 years (ref. 52) and wind turbines' lifespan follows the normal distribution with a mean value $\mu = 20$ years and a standard deviation of 0.25 μ (ref. 47).

Role of CE strategies in the demand for REEs

The REE inflows are calculated by multiplying the inflows of final products (EVs and wind power) and their corresponding REEs' material intensities. In the reduction strategy, the REEs' material intensities can be reduced, and in the substitution strategy, the REE-containing technologies can be substituted by the REE-free technologies. Therefore, the inflow of REEs is calculated as follows:

$$\text{inflow}_{i,t} = \text{inflow}_{i,t} \times \text{MI} \times (1 - \text{red}_t) \times (1 - \text{sub}_t) \quad (3)$$

where MI is the material intensity of a specific REE in its corresponding application (Supplementary Table 8), red_t is the reduction rate of the REE's material intensity as a result of the reduction strategy in time t and sub_t is the rate that the REE-containing technologies are replaced with the REE-free technologies as a result of the substitution strategy.

Notably, the mainstream technologies of wind power generators include direct-drive and geared-drive permanent magnet synchronous generators, which have different REE material intensities⁴. Therefore, the REE inflows in the wind power sector are calculated by the sum of two kinds of technologies. See more details in the Supplementary Information section 2.3.

As for the other sectors, we predict their annual demand for REEs assuming that it will grow at a compound annual growth rate of 3%, given the expected annual growth in global gross domestic product⁵³.

The outflows and in-use stocks of the REEs are calculated as follows:

$$\text{outflow}_{i,t} = \sum_{m < t} \text{inflow}_{i,m} \times L(t - m) \quad (4)$$

$$\text{stock}_{i,t} = \text{stock}_{i,0} + \sum_{m \leq t} \text{inflow}_{i,m} - \text{outflow}_{i,t} \quad (5)$$

where $\text{outflow}_{i,t}$ is the outflow of a specific REE in region i in time t , $\text{stock}_{i,t}$ is the in-use stocks of the specific REE and $\text{stock}_{i,0}$ is the initial in-use stocks of the REE. The lifespan of other sectors is assumed to follow the Weibull probability distribution with the most likely lifetime of ten years.

Role of CE strategies in the secondary supply of REEs

The secondary supply of REEs can be sourced from the reuse and recycling strategies. The reused REEs are the REEs contained in the EOL products with a reusable design that have not been reused before, which are calculated as equation (6).

$$\text{reused}_{i,t} = \sum_{m < t} (\text{inflow}_{i,m} - \text{reused}_{i,m}) \times L(t - m) \times \text{reusable_rate}_m \quad (6)$$

where $\text{inflow}_{i,m}$ is the REE inflows in time m and reusable_rate_m is the share of products with reusable design in time m .

It is assumed that if an EOL product is reusable, it will be reused preferentially; otherwise, it will be recycled. Therefore, the recycled REEs are calculated as equation (7).

$$\text{recycled}_{i,t} = (\text{outflow}_{i,t} - \text{reused}_{i,t}) \times \text{recycling_rate}_t \quad (7)$$

where $\text{outflow}_{i,t}$ is the outflow of a specific REE, and recycling_rate_t is the recycling rate of the REE in time t .

The secondary supply of REEs is then calculated as the sum of $\text{reused}_{i,t}$ and $\text{recycled}_{i,t}$.

The required reserves and losses of REEs

In consideration of the yield rate of REEs (60% in the extraction stage³⁰, 90% in the refining and separation stage and 98% in the fabrication stage⁵⁴), we define the required reserves (equation (8)) as the amount of REEs required to be mined from the in-ground stocks to satisfy their primary demand. The cumulative required reserves between 2021 and 2050 are calculated as equation (9), and the results are shown in Supplementary Tables 22–26. The losses of REEs here include the production losses and recycling losses, where the production losses are the amount of REEs lost in the process of refining, separation and fabrication (equation (10)), and the recycling losses are those REEs uncollected or lost in recycling process (equation (11)). The cumulative REE losses by time T are calculated as equation (12).

$$\text{RR}_{i,t} = (\text{inflow}_{i,t} - \text{reused}_{i,t} - \text{recycled}_{i,t}) / \text{py} \quad (8)$$

$$\text{CRR}_i = \sum_{t=2021}^{2050} \text{RR}_{i,t} \quad (9)$$

$$\text{pl}_{i,t} = \text{RR}_{i,t} \times (1 - \text{py}) \quad (10)$$

$$\text{rl}_{i,t} = \text{outflow}_{i,t} - \text{reused}_{i,t} - \text{recycled}_{i,t} \quad (11)$$

$$\text{CL}_{i,T} = \sum_{t \leq T} \text{pl}_{i,t} + \text{rl}_{i,t} \quad (12)$$

where $RR_{i,t}$ is the required reserves of region i in time t , is the full-cycle yield rate of REEs (which is 53%), CRR_i is the cumulative required reserves of region i , $pl_{i,t}$ is the production losses of region i in time t , $rl_{i,t}$ is the recycling losses of region i in time t and $CL_{i,T}$ is the cumulative (production and recycling) losses of region i by time T .

Primary supply of the REEs

On the basis of Roskill's forecast of RE production before 2030 in kiloton (kt) rare-earth oxide (REO)²⁸, we further assume a 5% compound annual growth rate of RE production after 2030 (Supplementary Fig. 7). Notably, RE production is compounded by various REOs, which need to be disaggregated into individual REEs. To do so, we estimate the average content of the REEs (indexed by k) in different regions (indexed by i) on the basis of equation (13).

$$REE_{i,k} = \sum_j reserve_share_{i,j} \times REO_{j,k} \times mf_k \quad (13)$$

where $REE_{i,k}$ is the average content of the REE k in region i , $reserve_share_{i,j}$ is the reserve share of mine j in region i , $REO_{j,k}$ is the content of the oxide of REE k in mine j ($j = 1, \dots, 20$) (Supplementary Table 16) and mf_k is the mass fraction of REE k in its oxide (85.7% for Nd, 82.8% for Pr, 87.1% for Dy and 89.2% for Tb). The result of the REEs' average content in different regions is shown in Supplementary Table 17. Then the primary supply of REEs in different regions can be calculated on the basis of their RE production (kt REO) and the corresponding REE average content (%), and the results are shown in Supplementary Table 18.

Reserves of the REEs

We consider two reserve scenarios in our analysis: present and dynamic reserves. The present reserves assume that there is no change in global RE reserves, which is measured by the RE reserves reported by the US Geological Survey (USGS) in 2022⁵⁵. The present reserves of different REEs can be calculated as the product of RE reserves reported by the USGS in kt REO and the corresponding REE average content (%). The present reserves of four REEs are shown in Supplementary Table 19. The dynamic reserves assume that RE reserves can change with new mines being explored. The historical RE reserves increase from 110 million tons (reported in 2012) to 120 million tons (reported in 2022) in the past decade, with a growth rate of 9% according to the USGS⁵⁵. We assume that the dynamic RE reserves will continue to grow at a rate of 9% per decade on the basis of present reserves.

Uncertainty and sensitivity analysis

Uncertainties of our analysis are related mainly to model assumptions about the lifespan of end uses and the REE material intensity in these end uses. Here we adopt the Monte Carlo simulation (100,000 iterations) to quantify the uncertainties of main model results, including annual regional RE demand and in-use stocks (Supplementary Figs. 34 and 35). In addition, sensitivity analysis for these key uncertainty factors is conducted (see details in Supplementary Information section 4.2), and the results are represented in Supplementary Fig. 33.

Overview of scenarios settings. Our work combines two series of scenarios in our DIRECCT modelling: climate target scenarios and circular economy scenarios. The details of scenario settings can be found in Supplementary Information section 1.2 and are summarized as follows.

Climate target scenarios. The climate goal scenarios talked about in this paper are the STEPS, SDS and NZE. STEPS reflects the current policy settings, SDS assumes that the world will reach net-zero emissions by 2070 (with many countries and regions reaching net zero much earlier) while NZE shows a pathway for the global energy sector to achieve net-zero CO₂ emissions by 2050^{56,57}. The detailed definitions and objectives of the climate goal scenarios are shown in Supplementary Table 1.

Circular economy scenarios. We discuss the effects of four individual CE scenarios—reduction, substitution, reuse and recycling—and the synergistic effect of their composite strategy. Substitution and reduction are considered the preventive measures, which aim to replace the REEs in final applications or decrease their intensity through improved designs, respectively. As the EOL measures, reuse aims to fulfil demand through reusable components or products and recycling involves the recovery of REEs for further use. In addition, a baseline scenario is set as a basis for comparison with these CE scenarios.

The narratives of the CE scenarios are as follows.

Reduction strategy. Reduction strategy involves reducing the material intensity of REEs through increasing material efficiency or optimizing product design⁵⁸. We assume a faster progress in HREE content reduction given the recent success in HREE reduction efforts²⁰.

Substitution strategy. Rare-earth permanent magnets (REPMs) can be substituted with REE-free magnets^{24,59}, where the EVs and wind turbines are assumed to have less potential to abandon the REPMs than other traditional sectors because REPMs have the advantages of compact size and high energy density that are favoured by the emerging sectors^{58,60}.

Reuse strategy. Reuse involves using an item again, usually for its original purpose or a similar one, without significant processing. Progressively more manufacturers will design and produce reusable REPMs, which are more implemented in EVs and wind turbines because they have large REPM sizes^{5,61,62}. The reusable REPMs can be reused only once.

Recycling strategy. Recycling involves breaking down used materials to create new products or raw materials. EVs and wind turbines have a higher recycling rate than other end uses because they are easier to trace and collect due to their large sizes and clear property rights.

Composite strategy. All four single CE strategies are implemented in parallel.

The key assumptions and parameters of these scenarios are shown in Supplementary Table 15.

Limitations

This study has several limitations. Our quantification of the in-use stocks of REEs is based on an implicit assumption that all the demand for REEs can be fulfilled by 2050. However, our study also indicates that the in-ground stocks of Dy and Tb could be exhausted by around 2035–2045, which means there are not enough REE in-ground stocks to fully meet their demand. Therefore, only the development of new mines or the secondary mining of tailings and scraps can be expected to support the in-use stocks landscape shown in Fig. 4b. However, it is important to highlight that this study does not encompass the intricate market dynamics between primary and secondary supplies of REEs, nor does it fully consider the interplay of market forces affecting the equilibrium of REE supply and demand. Further research has the potential to enhance the analysis of market-driven influences that govern the relationships among REE demand, primary supply and secondary supply.

Data availability

The data generated and/or analysed in this study are provided in supplementary information and in the figshare repository (<https://doi.org/10.6084/m9.figshare.24471670>). The published data sources underlying the parameters used in this study are documented in supplementary information. Source data are provided with this paper. Any additional data are available from the corresponding authors.

Code availability

The code used to manipulate the data and generate the results is available from the figshare repository (<https://doi.org/10.6084/m9.figshare.24471568>).

References

51. Baars, J., Rajaeifar, M. A. & Heidrich, O. Quo vadis MFA? Integrated material flow analysis to support material efficiency. *J. Ind. Ecol.* **26**, 1487–1503 (2022).
52. Fu, X. et al. Perspectives on cobalt supply through 2030 in the face of changing demand. *Environ. Sci. Technol.* **54**, 2985–2993 (2020).
53. Habib, K. & Wenzel, H. Exploring rare earths supply constraints for the emerging clean energy technologies and the role of recycling. *J. Clean. Prod.* **84**, 348–359 (2014).
54. Wang, Q.-C., Chen, W.-Q., Wang, P. & Dai, T. Illustrating the supply chain of dysprosium in China through material flow analysis. *Resour. Conserv. Recycl.* **184**, 106417 (2022).
55. *Mineral Commodity Summaries 2023* (USGS, 2023); <https://doi.org/10.3133/mcs2023>
56. *World Energy Outlook 2021* (IEA, 2021); <https://www.iea.org/reports/world-energy-outlook-2021>
57. *Global Energy and Climate Model* (IEA, 2022); <https://www.iea.org/reports/global-energy-and-climate-model>
58. Pavel, C. C. et al. Role of substitution in mitigating the supply pressure of rare earths in electric road transport applications. *Sustain. Mater. Technol.* **12**, 62–72 (2017).
59. Li, X.-Y., Ge, J.-P., Chen, W.-Q. & Wang, P. Scenarios of rare earth elements demand driven by automotive electrification in China: 2018–2030. *Resour. Conserv. Recycl.* **145**, 322–331 (2019).
60. Pavel, C. C. et al. Substitution strategies for reducing the use of rare earths in wind turbines. *Resour. Policy* **52**, 349–357 (2017).
61. Högberg S. et al. Direct reuse of rare earth permanent magnets—Wind turbine generator case study. *2016 XXII International Conference on Electrical Machines (ICEM)* 1625–1629 (IEEE, 2016).
62. Li, Z., Kedous-Lebouc, A., Dubus, J.-M., Garbuio, L. & Personnaz, S. Direct reuse strategies of rare earth permanent magnets for PM electrical machines—an overview study. *Eur. Phys. J. Appl. Phys.* **86**, 20901 (2019).

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Author contributions

P.W. and Y.-Y.Y. designed the research, with steering and supervision from W.-Q.C. and L.-H.C.; P.W., Y.-Y.Y. and O.H. contributed to the methodology; P.W., Y.-Y.Y., O.H., T.F. and L.-Y.C. conducted the analysis. P.W., Y.-Y.Y., O.H. and W.-Q.C. led the drafting of the manuscript. All authors contributed significantly to the final writing of the article.

Competing interests

The authors declare no competing interests.

Additional information

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